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DEPARTMENT OF TEACHER EDUCATION

SCHOOL OF EDUCATION AND LEADERSHIP

COLLEGES OF EDUCATION

END OF SEMESTER ONE EXAMINATIONS FOR LEVEL 300, 2023/2024

B.ED. PROGRAMME

COURSE CODE: **TEUP 305**

COURSE TITLE: **INTEGRATED SCIENCE III FOR UPPER PRIMARY**

Instruction: Answer **all** questions in **Section A** and any **three (3)** questions in **Section B**.

Time: 2 hours

SECTION A

[25 Marks]

Answer all the questions in this section.

1. Which of the following flowers contain both stamens and pistils?
   A. Bracteate flower
   B. Incomplete flower
   C. Perfect flower
   D. Staminate flower

2. The teeth are part of which system?
   A. Integumentary
   B. Nervous
   C. Skeletal
   D. A and B

3. During pollination, the pollen lands on which part of the flower?
   A. integument
   B. ovule
   C. stigma
   D. style

4. The flower is important to angiosperm plants because it helps in _____.
   A. pollination
   B. production of nectar
   C. production of seeds
   D. It gives off nice scent

5. If all floral whorls of a flower are present, it is a/an _____.
   A. androgynous
   B. complete flower
   C. hermaphrodites
   D. incomplete flower

6. The brightly coloured part which attracts insects to the flower is known as _____.
   A. corolla
   B. pistil
   C. sepals
   D. stamens

7. A typical flower has the calyx, corolla, androecium, and the _____.
   A. anther
   B. gynoecium
   C. petal
   D. sepal

8. The outermost whorl of the flower which protects the unopened bud is known as _____.
   A. corolla
   B. pistil
   C. sepal
   D. stamens

9. A moving air is often regarded as _____.
   A. energy
   B. force
   C. pressure
   D. wind

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10. The skeletal system relies on the... to remove waste produced by bone cells.
    A. digestive system
    B. endocrine system
    C. respiratory system
    D. urinary system

11. The cardiovascular system includes the following **except** _____.
    A. Blood
    B. Blood vessels
    C. Heart
    D. Skin

12. The most abundant element in the air is oxygen.
    A. False
    B. True

13. Air protects the earth from the impact of harmful radiations like UV rays and celestial particles like cosmic rays, and asteroids.
    A. False
    B. True

14. Plants with male and female flowers on separate plants are termed dioecious.
    A. False
    B. True

15. The ovary is the swollen base of the carpel where ovules are found.
    A. False
    B. True

16. Which of the following seeds are all dispersed by wind except _____.
    A. Silk cotton
    B. Tridax
    C. Tecoma
    D. Desmodium

17. To conserve water effectively, what actions should **NOT** be taken?
    A. Turn off the taps and repair leaking pipes.
    B. Collect rainwater.
    C. Never throw garbage, wash clothes & bathe animals in rivers.
    D. Irrigate the rivers to areas with no water.

18. Why is water known as a universal solvent?
    A. It has many uses.
    B. It can remove all kinds of dirt.
    C. It can dissolve many substances.
    D. It acts as a medium for chemical reactions in the body.

19. What change in state occurs when thermal energy is increased in a solid?
    A. Freezing
    B. Melting
    C. Evaporation
    D. Condensation

20. The part of the earth which contains water is called the_____.
    A. Atmosphere.
    B. Lithosphere.
    C. Hydrosphere.
    D. Troposphere.

21. Identify the four whorls of a flower: the corolla, androecium, gynaecium, and _____.
22. Break down the components of the carpel, which include the ovary, style, and _____.
23. The process of pollen tube growth towards the ovary, passing through a tiny hole is called _____.
24. Evaluate different types of flowers based on their structure: Flowers containing only an androecium are known as _____.
25. Evaluate different types of flowers based on their structure: A flower devoid of a stalk and borne directly on the stem is called _____.

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SECTION B – (75MARKS)

Answer any three (3) questions from this section. All questions carry equal marks.

1. a) Differentiate between true fruit and false fruit. **[4 marks]**

b) Explain the adaptations of seeds/fruits dispersed by:

i. Wind, using an example.
ii. Animals, using an example.
iii. Water, using an example. **[9 marks]**

c) Provide **one** (1) example of seed dispersed by an explosive mechanism. **[2 marks]**

d) Plan a teaching approach for explaining the parts of a flower to Basic 5 pupils. **[9 marks]**

2. a) Define the following terms as applied to fruits:

i. drupe,
ii. berry,
iii. legume/pod,
iv. achene,
v. false fruit. **[10 marks]**

b) Explain the concept of germination. **[2 marks]**

c) Name **two** (2) types of germination and describe **each**. **[4 marks]**

d) Provide **two** (2) examples each of seeds that undergo **each** type of germination. **[4 marks]**

e) Classify the following fruits as Multiple, Aggregate, or True fruits: orange, strawberry, tomato, sweet apple, pineapple. **[5 marks]**

3. a) Recall and match the parts of a flower in the left-hand column to their correct functions in the right-hand column. **[10 marks]**

Parts of a flower	Function
Receptacle	Supports the anther
Style	Produces pollen
Stigma	Makes a sugary liquid
Pollen	Receives pollen
Petal	Contains the male gametes
Nectar	Attracts insects
Sepal	Forms the base of the flower
Anther	Holds up the stigma
Ovary	Protects the flower in the bud
Filament	Contains the female gametes

b) Describe the process of fertilization in flowering plants. **[7 marks]**

c) State **two** (2) differences between a fruit and a seed. **[4 marks]**

d) Define what constitutes a complete flower. **[2 marks]**

e) Mention **two** (2) uses of a flower. **[2 marks]**

4. a) Draw the root system of a flower plant and label its developmental parts.

[10 marks]

b) Identify and explain which part of the root undergoes elongation during growth.

[3 marks]

c) State **four (4)** science curriculum materials you observed during your STS scheduled this semester in your partner schools. **[2 marks]**

d) As a teacher to teach Basic six pupils science, prepare a checklist of **five (5)** materials and explain how you will employ them to teach classification of plants.

[10 marks]

5. a) Explain the distinction between a scheme of work and a lesson plan. **[10 marks]**

b) Explain the following terms in relation to the lesson plan:

i. Learning indicator
ii. Performance indicator
iii. Strands
iv. Sub-strands
v. Teaching/Learning Resources **[10 marks]**

c) Copy and complete the unique annotation "B3.2.4.1.2" in the table provided as used for numbering the learning indicators in the curriculum for referencing purposes.

[5 marks]

ANNOTATION	MEANING/ REPRESENTATION
B3	
2	
4	
1	
2	